PRECALCULUS · UNIT 8: PROBABILITY

Sports Draft Analytics

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students run a basketball team's analytics. Lineups use permutations and combinations, a draft lottery uses conditional probability, scouting data uses two-way tables, free throws follow a binomial distribution, and trades are judged by expected value.

The decisions are realistic and the answers are surprising enough to spark discussion.

Standards Addressed

- S-CP.A.1–5 Independence and conditional probability; two-way tables
- S-CP.B.6–9 Addition and multiplication rules; permutations and combinations
- S-MD.A.1–4 Random variables, expected value
- S-MD.B.5–7 Use probability to evaluate decisions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a draft lottery video. Part 1 counting.
Day 2	Part 2 conditional probability.
Day 3	Part 3 binomial, expected value, and the report.

Materials

- Student handout and Scouting Data Cards
- Graphing calculators (nCr, binompdf)
- Optional: real NBA lottery odds

How the Three Versions Differ

Level	What changes
CP2	Counting principle, permutations vs. combinations, two-way table probabilities, and expected value.
CP1	Lineup probabilities, independence test, lottery top-2, binomial probabilities, and trade comparison.
Honors	Complements, case counting, Bayes-style scouting test, top-3 lottery, binomial σ, and variance of trades.

Watch For (Common Misconceptions)

- Using permutations when order does not matter.
- Confusing P(A | B) with P(B | A).
- Forgetting the ${}_nC_k$ factor in binomial probabilities.

Extension Ideas

- Simulate free throws with a random number generator.
- Analyze real team statistics.
- Design a fair lottery for a school event.

Answer Key – CP2

Part 1: Count the Options

#	Task	Answer
1	How many two-digit jersey numbers with no repeated digit are possible (first digit can be 0)?	$10 \times 9 = 90$
2	In how many ways can the coach choose 5 starters from 13 players (order does not matter)?	${}_{13}C_5 = 1{,}287$
3	In how many ways can the coach choose a captain and a co-captain from 13 players?	${}_{13}P_2 = 156$
4	Explain why the captains question is a permutation and the starters question is a combination.	Captain and co-captain are different roles (order matters); starters are a group.

Part 2: Draft Data

#	Task	Answer
5	Make a two-way table: guards/others and shooters/non-shooters.	Guards: 24 shooters, 36 non; others: 35 shooters, 105 non; totals 59, 141, 200
6	What is P(shooter)?	$\frac{59}{200} = 29.5\%$
7	What is P(shooter | guard)?	$\frac{24}{60} = 40\%$
8	What is the probability that the Comets win the top pick in the lottery?	$\frac{140}{1{,}000} = 14\%$

Part 3: Expected Value

#	Task	Answer
9	Find the expected extra wins for Trade A.	$0.6(8) + 0.4(2) = 5.6$ wins
10	Find the expected extra wins for Trade B.	$0.35(12) + 0.65(-1) = 3.55$ wins
11	Which trade would you recommend? Why?	Trade A; it has a higher expected value and no chance of losing wins.

Reflection

#	Prompt	Look for
R1	How is P(shooter | guard) different from P(shooter)?	It only looks at guards, so the denominator is 60 instead of 200.
R2	What does expected value tell the owner?	The average result if the decision could be repeated many times.

Answer Key – CP1

Part 1: Lineups

#	Task	Answer
1	How many starting lineups have 2 guards, 2 forwards, and 1 center?	${}_5C_2 \times {}_5C_2 \times {}_3C_1 = 10 \times 10 \times 3 = 300$
2	If 5 players are chosen at random, what is the probability of getting exactly that mix?	$\frac{300}{1{,}287} \approx 0.233$
3	In how many orders can the 5 starters be introduced?	5! = 120
4	What is the probability that a random 5-player group includes the star and the backup point guard?	$\frac{{}_{11}C_3}{{}_{13}C_5} = \frac{165}{1{,}287} \approx 0.128$

Part 2: Conditional Probability

#	Task	Answer
5	Find P(guard | shooter).	$\frac{24}{59} \approx 0.407$
6	Are "guard" and "shooter" independent? Compare P(shooter | guard) with P(shooter).	0.40 vs. 0.295; not independent
7	The top pick is 14%. If the Comets miss the top pick, they have 134 of the remaining 860 combinations for the second pick. Find P(top 2).	$0.14 + 0.86 \times \frac{134}{860} = 0.14 + 0.134 = 0.274$
8	Find P(not in the top 2).	0.726

Part 3: Free Throws & Trades

#	Task	Answer
9	Find the probability the star makes exactly 8 of 10 free throws.	${}_{10}C_8\,(0.8)^8(0.2)^2 \approx 0.302$
10	Find the probability of making at least 9 of 10.	${}_{10}C_9(0.8)^9(0.2) + 0.8^{10} \approx 0.268 + 0.107 = 0.376$
11	Compute the expected extra wins for Trades A and B and the standard of risk (chance of losing wins). Recommend one.	A: 5.6, no loss; B: 3.55, 65% chance of −1. Trade A.

Reflection

#	Prompt	Look for
R1	How do you test whether two events are independent?	Check if P(A | B) = P(A) (or P(A and B) = P(A)P(B)).

#	Prompt	Look for
R2	Why is the binomial distribution a good model for free throws?	A fixed number of independent shots with the same probability of success.

Answer Key – Honors

Part 1: Lineup Probability

#	Task	Answer
1	How many 5-player lineups have at least one center?	${}_{13}C_5 - {}_{10}C_5 = 1{,}287 - 252 = 1{,}035$
2	What is the probability a random lineup has more guards than forwards?	Cases (g, f, c) with g > f: (2,0,3) 10, (2,1,2) 150, (3,0,2) 30, (3,1,1) 150, (3,2,0) 100, (4,0,1) 15, (4,1,0) 25, (5,0,0) 1 → 481; $\frac{481}{1{,}287} \approx 0.374$
3	The 5 starters line up for photos with the 2 tallest at the ends. How many arrangements?	$2! \times 3! = 12$
4	Explain how you used complements in the first question.	Counting lineups with no centers and subtracting from all lineups is easier than counting 1, 2, or 3 centers.

Part 2: Scouting with Conditional Probability

#	Task	Answer
5	A prospect is a shooter. Find P(guard | shooter) and P(other | shooter).	$\frac{24}{59} \approx 0.407$ and $\frac{35}{59} \approx 0.593$
6	A scouting test flags 90% of shooters but also flags 10% of non-shooters. If a prospect is flagged, what is the probability they are a shooter?	Flagged shooters $0.9 \times 59 = 53.1$; flagged non-shooters $0.1 \times 141 = 14.1$ → $\frac{53.1}{67.2} \approx 0.79$
7	Find P(Comets get a top-3 pick) if the second-pick chance is 134 of the remaining 860 and the third-pick chance is 127 of the remaining combinations after two picks (assume 726 remain).	$0.14 + 0.86(\frac{134}{860}) + 0.726(\frac{127}{726}) = 0.14 + 0.134 + 0.127 = 0.401$
8	Explain why the second-pick probability depends on the first pick.	Once the top pick's combinations are removed, the remaining combinations change the denominator (dependent events).

Part 3: Decisions

#	Task	Answer
9	Find the expected number and standard deviation of made free throws in 10 attempts.	$\mu = 8$; $\sigma = \sqrt{(10 \times 0.8 \times 0.2)} \approx 1.26$
10	Find the variance of extra wins for Trade A and Trade B. Which is riskier?	A: $E = 5.6$, $E(X^2) = 0.6(64) + 0.4(4) = 40$ → Var = 8.64. B: $E = 3.55$, $E(X^2) = 0.35(144) + 0.65(1) = 51.05$ → Var ≈ 38.45. B is riskier.
11	Write the Draft Room Report with recommendations backed by counting, conditional probability, and expected value.	Answers vary.

Reflection

#	Prompt	Look for
R1	Why can a test that is 90% accurate still produce many false flags?	Non-shooters greatly outnumber shooters, so 10% of them is a large count.
R2	Why should decisions consider variance as well as expected value?	Two options with similar averages can have very different risks.